
🌐 AWS NAT Instance – Detailed Overview + Use Case

🎯 What Is a NAT Instance?

A **NAT instance** is an **EC2 instance** configured to **allow private subnet resources (like EC2s)** to **access the internet**, while still **blocking inbound connections from the internet**.

> ✅ Outbound internet access
> ❌ Inbound internet access (unless explicitly allowed)

🏗️ When to Use NAT Instance?

- You want to **enable internet access for EC2s in private subnets**
- You need **custom control** (firewall rules, logging, special AMIs)
- You're working in **low-traffic** or **testing environments**

> 🔁 For production, **NAT Gateway** is recommended (scalable, managed).

🔍 NAT Instance vs NAT Gateway

Feature	NAT Instance	NAT Gateway
Type	EC2 instance	Managed service

High availability	❌ Manual setup (1 AZ only)	✅ Multi-AZ support
Bandwidth	Limited by instance type	Up to 45 Gbps
Resiliency	Manual failover required	Auto failover
Cost	✅ Cheaper for small setups	❌ More expensive
Logging/Customization	✅ Full control	❌ No direct customization

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📦 Requirements

- **One NAT instance** in a **public subnet**
- Elastic IP attached to NAT instance
- **Source/Destination Check: Disabled**
- Route table for **private subnet** should route 0.0.0.0/0 → NAT instance

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🔐 Security Groups

- NAT instance should allow:
  - **Inbound**: Only from private subnet CIDR range (for HTTP/HTTPS)
  - **Outbound**: Open to the internet (80, 443)

🛠️ Step-by-Step Lab: NAT Instance Setup

🧱 Architecture

[Private Subnet EC2] ───▶ [NAT Instance in Public Subnet] ───▶ [Internet]

1️⃣ VPC and Subnets

Create a VPC:

- CIDR: 10.0.0.0/16

Create subnets:

- Public: 10.0.1.0/24
- Private: 10.0.2.0/24

Create Internet Gateway, attach to VPC.

2️⃣ Launch NAT Instance

1. Go to **EC2** → **Launch instance**
2. Choose NAT AMI:
   - **Amazon NAT AMI**: Search for *"amzn-ami-vpc-nat"* in community AMIs
3. Select **t3.micro** or larger

4. Subnet: Public Subnet
    - **Enable auto-assign public IP**
5. Attach **Elastic IP** (optional but recommended)
6. **Disable Source/Destination Check**:
    - Go to **EC2** → **Actions** → **Networking** → **Change Source/Dest Check**
    - Choose "Disabled"

3️⃣ Configure Security Group

Create a **NAT-SG** with:

- Inbound:
    - From 10.0.2.0/24, allow ports 80 (HTTP), 443 (HTTPS)
- Outbound:
    - Allow all (0.0.0.0/0)

4️⃣ Launch EC2 in Private Subnet

1. Launch a Linux EC2 in **Private Subnet**
2. No public IP

3. Attach security group to allow:
    - Outbound: HTTP, HTTPS
    - SSH from bastion (optional)

5️⃣ Update Route Table

Route table for **Private Subnet**:

- Add route:
    - 0.0.0.0/0 → Target: **NAT Instance ID**

6️⃣ Test Internet Access

From Private EC2:

curl https://google.com

✅ If setup correctly, request will pass through NAT instance and reach the internet.

⚠️ Common Troubleshooting

Problem	Solution
❌ No internet access	Check route table of private subnet
❌ Instance unreachable	Disable Source/Dest Check on NAT

❌ DNS issues	Ensure correct DNS server (AmazonProvidedDNS)
❌ HTTP works, HTTPS fails	Check SG rules on NAT allow 443

🧼 Cleanup

- Terminate NAT and EC2 instances
- Delete VPC and subnets
- Release Elastic IP (if used)

✅ Summary

Component	Role
NAT Instance	Provides outbound internet for private subnets
Public Subnet	Hosts NAT instance
Private Subnet	Hosts workloads needing internet
Route Table	Routes private subnet via NAT
